

Aligning Amino Acid Sequences: Comparison of Commonly Used Methods

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Summary. We examined two extensive families of protein sequences using four different alignment schemes that employ various degrees of “weighting” in order to determine which approach is most sensitive in establishing relationships. All alignments used a similarity approach based on a general algorithm devised by Needleman and Wunsch. The approaches included a simple program, UM (unitary matrix), whereby only identities are scored; a scheme in which the genetic code is used as a basis for weighting (GC); another that employs a matrix based on structural similarity of amino acids taken together with the genetic basis of mutation (SG); and a fourth that uses the empirical log-odds matrix (LOM) developed by Dayhoff on the basis of observed amino acid replacements. The two sequence families examined were (a) nine different globins and (b) nine different tyrosine kinase-like proteins. It was assumed *a priori* that all members of a family share common ancestry. In cases where two sequences were more than 30% identical, alignments by all four methods were almost always the same. In cases where the percentage identity was less than 20%, however, there were often significant differences in the alignments. On the average, the Dayhoff LOM approach was the most effective in verifying distant relationships, as judged by an empirical “jumbling test.” This was not universally the case, however, and in some instances the simple UM was actually as good or better. Trees constructed on the basis of the various alignments differed with regard to their limb lengths, but had essentially the same branching orders. We suggest some reasons for the different effectivenesses of the four approaches in the two different sequence settings, and offer some rules of thumb for assessing the significance of sequence relationships.

Key words: Amino acid sequence alignment — Tyrosine kinases — Globins — Homologies

Introduction

Numerous articles have been written on the subject of aligning protein amino acid sequences, and the advantages of different approaches have been extolled by various authors (e.g., Fitch 1966; Haber and Koshland 1970; McLachlan 1971; Sellers 1974; Waterman et al. 1976; Dayhoff 1978; Smith et al. 1981; Gotoh 1982; Dayhoff et al. 1983). For the most part, these authors have not conducted rigorous comparisons of large numbers of real proteins to assess the actual benefits of a given procedure. In this article we apply four different alignment schemes to two large families of sequences and compare these schemes with regard to the nature of the alignments they generate and their sensitivity in detecting distant relationships. The parameters and factors chosen for study relate to the relative importance of giving weight to nonidentical amino acids that are structurally similar or that are genetically possible as results of single-base substitutions.

All of the methods explored in this study were examined using procedures in which maximization of similarity is the goal. As such, the programs are based on the general approach of Needleman and Wunsch (1970). This algorithm obtains a maximum score based on rewards for similarities or identities between two sequences and penalties for deletions in either sequence. The deletions are commonly referred to as “gaps.” In the present study gaps have not been weighted by length, nor have overhangs (terminal gaps) been counted. In a separate study (D.F. Feng, M.S. Johnson, and R.F. Doolittle, manuscript in preparation), we have explored the fundamental difference between similarity schemes

and difference metrics, as well as the attributes of length-dependent penalties for gaps and overhangs.

Methods

All studies were done on a DEC 11/750 VAX computer that is part of the Chemistry Department Computer Center, University of California, San Diego. The UNIX operating system was used exclusively; all programs were written in the C programming language (Kernighan and Ritchie 1978).

Needleman and Wunsch Algorithm

The dynamic programming for the similarity programs was based on the algorithm first introduced by Needleman and Wunsch (1970). Briefly, a rectangular matrix is established, the size of which is determined by the lengths of the two sequences being compared. The upper left-hand and lower right-hand corners of the matrix correspond to the amino and carboxy terminals of the sequences, respectively. In the simplest case (unitary matrix), a value of 1 is placed in the matrix whenever an amino acid in one of the sequences is identical to an amino acid in the other sequence, regardless of position. All nonidentical amino acids are assigned values of zero. Starting from the carboxy-terminal corner of this matrix and proceeding row by row to the amino terminal, the scores are added; as a result this matrix is transformed into a cumulative similarity score matrix. This means that any matrix element, say a_{ij} , is the similarity score obtained by aligning these two sequences starting from the i -th residue in one sequence and j -th residue in the other, both ending at the carboxy terminals. The maximum value along the amino-terminal portion of the matrix is the maximum similarity score corresponding to the alignment of the two full sequences. Any departure from the diagonal is equivalent to inserting a gap in one or the other sequence. Accordingly, a penalty has to be imposed in each such instance. The algebraic sum of the cumulative score and the penalties is used in outlining the best path.

This algorithm is readily adapted for weighting schemes merely by including a table in which each of the 190 amino acid interchanges is given an appropriate value. Instead of being limited to values of 0 and 1 for nonidentical and identical residues, the system can now give weight to similar nonidentical residues. The best set of matches—the highest cumulative scores including gap penalties—will then correspond to the best alignment for this scheme.

Methods for Determining the Significance of Alignment Scores

The significance of binary alignments was assessed by comparing the similarity score of the authentic comparison with the mean similarity scores determined for pairs of randomized ("jumbled") sequences of the same length and composition. If the authentic alignment score is more than +3.0 standard deviations (SD) above the mean of the randomized comparisons, the odds of obtaining the alignment by chance are less than one in a thousand, assuming a normal distribution of randomized scores. A value of +3.0 SD is often used as a cutoff point for establishing common ancestry in binary comparisons (Dayhoff 1978), although it should be mentioned that in the absence of independent evidence a cutoff of +5.0 SD is sometimes used (Dayhoff et al. 1983).

To answer the question of how many randomized comparisons are necessary to make these assessments, we compared four different pairs of sequences and varied the number of randomization alignments to see what impact it would have on the final

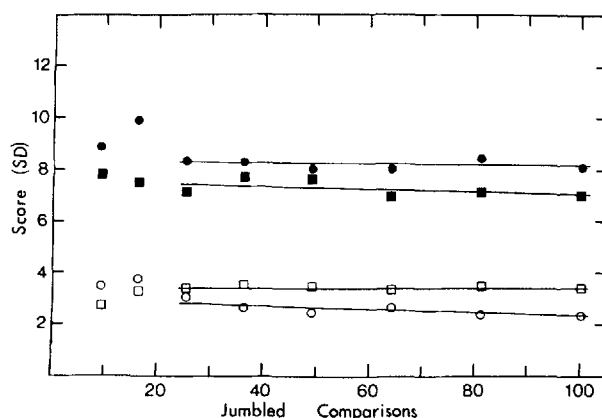


Fig. 1. Effect of the number of "jumbled" comparisons on the statistical significance of alignment scores. The following pairs of sequences were aligned and the sequences subjected to increasing numbers of jumbled comparisons: ●, v-src/cadk; ■, hahu/heha; □, heha/haew; ○, v-raf/cadk. The significance of the results in terms of standard deviation was then determined (see text). All comparisons were made with the log-odds matrix (LOM) approach. Only the data for 25 or more jumbled comparisons were used in finding the best line fit by linear regression. (For definitions of sequences, see Tables 3 and 4; cadk sequence used here is 18 residues longer)

assessment (see Fig. 1). Comparisons of two sets of sequences that are only marginally related and two comparisons that yielded quite significant alignment scores both indicated that fewer than 25 comparisons are needed to achieve a genuinely reflective score. In studies reported in this article, 100 jumbled comparisons were conducted in testing each pair of globin sequences; for the tyrosine kinase-type sequences, which on the average are twice the length of those of the globins, only 36 jumbled comparisons were made each time.

Definitions

In this study the percentage identity is defined as the number of identities observed after the alignment of the two sequences divided by the number of residues in the shorter sequence. The relative evolutionary distance was calculated by taking the negative log of the adjusted percent identity. Normalized alignment scores (NAS) are merely the alignment scores obtained in a given comparison divided by the number of residues in the shorter sequence.

The Four Scoring Approaches

1) *Unitary Matrix (UM)*. The UM scoring approach focuses on residues that are identical. When identical residues are encountered in the two sequences being compared, a matching score of 1.0 is awarded, except for matched cysteines, for which the score is 2.0 (Doolittle 1981). When two different residues are aligned, a mismatching score of 0.0 is assigned. A gap penalty of -2.5 is incorporated when a gap, regardless of size, appears in the alignment. This penalty value was determined empirically by performing a large number of different alignments (Doolittle 1981).

2) *Genetic Code Matrix (GC)*. The scoring system used in this system is complementary to the three-base-change values that have been used with difference measures (Sellers 1974; Smith et al. 1981). The minimum number of base changes needed to convert any residue to itself is naturally zero. Similarly, two mis-

Table 1. Matrix values used for the SG scoring system (upper triangle) and Dayhoff's (1978) log-odds scoring system (lower triangle; log-odd scores are multiplied by 10)

	C	S	T	P	A	G	N	D	E	Q	H	R	K	M	I	L	V	F	Y	W
C	6	4	2	2	2	3	2	1	0	1	2	2	0	2	2	2	2	3	3	3
S	20	6	5	4	5	5	5	3	3	3	3	3	3	1	2	2	2	3	3	2
T	8	10	6	4	5	3	2	2	3	3	3	3	2	2	2	3	3	2	2	1
P	6	9	11	6	5	3	4	4	3	2	2	3	2	1	2	2	5	2	2	2
A	5	9	8	14	6	3	4	4	2	1	3	2	1	2	2	2	4	1	2	3
G	6	9	9	9	10	6	5	3	3	4	2	4	1	2	1	2	1	3	0	N
N	5	9	8	7	9	13	6	5	4	3	2	3	0	1	1	1	3	1	2	0
D	4	9	8	7	8	8	10	6	4	2	2	4	1	1	1	1	4	0	1	1
E	3	8	8	7	8	8	9	11	12	6	4	3	4	2	1	2	2	1	2	1
Q	3	7	7	8	8	7	9	10	10	6	4	3	4	2	1	3	1	2	3	1
H	5	7	7	8	7	6	10	9	9	11	14	6	5	2	2	2	2	1	1	2
R	4	8	7	8	6	5	8	7	7	9	10	14	6	2	2	2	3	0	1	1
K	3	8	8	7	7	6	9	8	8	9	8	11	13	6	4	5	4	2	2	3
M	3	6	7	6	7	5	6	5	6	7	6	8	8	14	6	5	5	4	3	2
I	6	7	8	6	7	5	6	6	6	6	6	6	6	10	13	6	4	3	3	4
L	2	5	6	5	6	4	5	4	5	6	6	5	5	12	10	14	6	5	3	V
V	6	7	8	7	8	7	6	6	6	6	6	6	6	10	12	10	12	6	5	3
F	4	5	5	3	4	3	4	2	3	3	6	4	3	8	9	10	7	17	6	W
Y	8	5	5	3	5	3	6	4	4	4	8	4	4	6	7	7	6	15	18	
W	0	6	3	2	2	1	4	1	1	3	5	10	5	4	3	6	2	8	8	25

Table 2. Attempt to reproduce a series of eight comparisons originally reported by Dayhoff (1978)^a

Comparison ^b	UM		LOM		GC		SG	AAAM
	This study	Dayhoff	This study	Dayhoff	This study	Dayhoff	This study	Dayhoff
xsm/zns	1.5	3.1	1.9	2.9	2.7	3.2	2.5	2.6
fecl/fesg	0.9	0.1	3.8	3.4	2.2	1.6	4.0	1.8
hahu/myhu	12.4	5.8	12.7	10.7	11.4	6.6	15.7	9.9
hahu/gice	0.3	2.0	2.6	3.5	2.2	2.4	3.0	3.2
ccho/ccsg	5.7	4.5	6.1	6.1	5.2	4.3	4.7	7.3
ccho/ccdv	0.5	0.2	3.8	3.9	0.0	0.4	0.2	0.4
mghu/m3hu	3.2	3.6	3.4	4.8	3.5	3.3	4.0	4.7
m3hu/hund	6.8	4.7	14.0	12.1	9.6	9.0	8.6	9.2
Mean	3.9	3.0	6.0	5.9	4.6	3.9	5.3	4.9

^a The statistical significances of comparisons are expressed in terms of standard deviations. The comparisons reproduced in this study are based on 36 randomized comparisons; those of Dayhoff (1978) are based on 100 jumbled comparisons for the UM and GC approaches and 300 jumbles in the LOM case. Three hundred jumbles were also employed by Dayhoff in comparisons based on the "alternative amino acid matrix" (AAAM) developed by McLachlan (1971).

^b The codes for the sequences that were compared are: xsm, antibacterial substance A, *Streptomyces carzinostaticus* F41; zns, neocarzinostatin, *S. carzinostaticus* F41; fecl, ferredoxin, *Clostridium pasteurianum*; fesg, ferredoxin, *Spirulina maxima*; hahu, α -hemoglobin, human; myhu, myoglobin, human; gice, globin CTT-III, midge larva; ccho, cytochrome C, horse; ccsg, cytochrome C6, *Spirulina maxima*; cc dv, cytochrome C553, *Desulfovibrio vulgaris*; mghu, beta 2 microglobulin, human; m3hu, immunoglobulin μ -chain C4 homology region, human myeloma protein designated Gal; hund, immunoglobulin ϵ -chain C4 homology region, human myeloma protein designated Nd.

matched (nonidentical) amino acid residues can be made equivalent by one, two, or three base changes. Ordinarily the GC approach is most often used with difference measures; it has the advantage that the results can be applied directly to the construction of phylogenetic trees. To make this method appropriate for use in a similarity measure, the values were initially transformed such that zero, one, two, and three base changes were given scores of 3, 2, 1, and 0, respectively, as had been done in a similar exercise by Dayhoff (1978). The gap penalty in these cases was -3, a value determined empirically by applying penalties ranging from -1 to -5 in a large number of comparisons.

We soon realized, however, that the Dayhoff adaptation of the GC approach was flawed, in that a 3-2-1-0 weighting scheme is not reciprocally related, in a strict sense, to a 0-1-2-3 base-substitution scheme. Accordingly, we revised those weights to a 4-2-1-0 system and changed the gap penalty to -4. The performance of the GC method was significantly improved by these changes.

3) *Structure-Genetic (SG) Matrix.* The SG matrix scoring system takes into account the structural similarities of amino acids, as well as the likelihoods of interchanges. Its rationale is the fact

that more than half of the allowed 75 single-base changes lead to structurally dissimilar amino acids (Doolittle 1979). A similar attempt at such an approach to weighting was undertaken by McLachlan (1972). We determined an appropriate gap penalty empirically, as described for the GC approach; this value was -5.5. The assigned scores for the 190 amino acid interchanges, which range from 0 to 6, are given in Table 1.

4) *Dayhoff's Log-Odds Matrix (LOM)*. Dayhoff (1972, 1978) made an extensive semiempirical model study on evolutionary changes in proteins. Initially she selected 71 groups of closely related proteins, aligned them, and constructed phylogenetic trees. They inferred ancestral sequences from the nodes of the trees. Based on the inferred ancestral sequences, she counted the actual number of i-type amino acids that had been replaced by j-type amino acids, A_{ij} , during the course of evolution, PAMs (accepted point mutations). From these changes, she determined the relative mutability, m_i , of each of the 20 amino acids. Combining A_{ij} and m_i yielded the elements of the mutation-probability matrix. When this matrix was used in a Monte Carlo setting for changing a particular protein sequence, the evolutionary history of the sequence could be charted with regard to both the actual number of mutations that had occurred and the distance between the mutated and initial sequences.

Using the mutation-probability matrix, Dayhoff (1972, 1978) devised a scoring system for detecting the relatedness of a pair of proteins: the so called log-odds matrix (LOM). The LOM was the fourth system tested in our study. The actual weights in her table ranged from -0.8 to 1.7. These values and a gap penalty of 0.0 were transformed for use in our similarity alignment approach by the addition of 0.8 to each member in order to obtain a wholly positive scoring system (Table 1).

Other Comparisons

In an early comparison of various alignment schemes, Dayhoff (1978) compared several proteins by four procedures, including a UM approach, the genetic code, a SG scheme (McLachlan 1971), and their own LOM procedure. Although we were not satisfied that the sequences chosen were sufficiently representative to yield a fair test, we felt it important to try to reexecute the same comparisons to see if our programs were accurately reproducing her schemes. In general, reasonably similar results were obtained, although the GC approach improved because of the stronger weight given to identities. The UM approach also scored better, although the reasons for this are not immediately apparent (Table 2); tests revealed that the difference was not due to the slightly different gap penalties employed.

Construction of Phylogenetic Trees

Phylogenetic trees are constructed most directly from difference measures. Consequently, we transformed the similarity scores to difference scores by means of the relationship

$$D = -\ln(S_{\text{eff}}) \times 100, \quad (1)$$

in which the effective similarity score S_{eff} is defined as follows:

$$S_{\text{eff}} = \frac{S_{\text{real}} - S_{\text{rand}}}{S_{\text{iden}} - S_{\text{rand}}}. \quad (2)$$

S_{real} is the maximum similarity score calculated for a pair of authentic protein sequences, S_{rand} is the mean of the similarity scores obtained when sequences of the same length and composition are scrambled and aligned, and S_{iden} is the average authentic score obtained when each of the two sequences of interest is aligned with itself.

Table 3. The nine globin sequences compared in this study

Designation	Source	Sequence length (residues) ^a
hahu	α -Hemoglobin, human	141
hbhu	β -Hemoglobin, human	146
hghu	γ -Hemoglobin, human	146
myhu	Myoglobin, human	153
heha	Hemoglobin, hagfish (<i>Myxine glutinosa</i>) (Liljeqvist et al. 1979)	148
mycr	Myoglobin, gastropod (<i>Cerithidea rhizophorarum</i>) (Takagi et al. 1983)	151
hety	Hemoglobin, earthworm (<i>Tylorhynchus heterochaetus</i>) (Suzuki et al. 1982)	139
haew	Hemoglobin, earthworm (<i>Lumbricus terrestris</i>) (Garlick and Riggs 1982)	157
gpfb	Leghemoglobin, kidney bean (<i>Phaseolus vulgaris</i>)	145

^a The full lengths of these sequences were used for comparisons

A difference matrix can then be constructed with the D values, and an initial phylogenetic tree can be derived as described by Fitch and Margoliash (1967). The corresponding percentage standard deviation (PSD) between the difference matrix obtained from the alignments and another difference matrix obtained from the tree limb lengths, PSD_0 , was computed. Biologically reasonable alternatives to the original tree were then considered and a new PSD, PSD_1 , determined. The new tree was deemed acceptable only if $PSD_1 < PSD_0$. The trees were subsequently examined by the minimal tree length method (Fitch and Smith 1982); in all cases the results were consistent with those obtained by the PSD procedure.

Sequences Used for Comparisons

Two groups of sequences were used in this study. One of these was made up of nine hemoglobin and myoglobin sequences (Table 3); three human hemoglobins (α -, β -, and γ -chains), human myoglobin, hagfish hemoglobin, a gastropod myoglobin, two earthworm hemoglobins, and the kidney bean leghemoglobin. The globin sequences were taken from the 1978 *Atlas of Protein Sequence and Structure* (Dayhoff 1978) and NEWAT (Doolittle 1981). The latter currently contains about 1000 sequences taken from the original literature from 1978-1984.

The second group of sequences studied consisted of nine tyrosine kinase-like sequences: seven tyrosine kinase-like oncogenes, a yeast cell division cycle protein (*cdc28*), and bovine cyclic-AMP-dependent protein kinase (*cadk*). Both of these sequences have been reported to be homologous to the tyrosine kinase family, *cdc28* by Lorincz and Reed (1984) and the *cadk* by Barker and Dayhoff (1982). The lengths of the sequences and the portions of each sequence used in comparisons are given in Table 4 and Fig. 2. All of the tyrosine kinase-like sequences are listed in the 1984 version of NEWAT (Doolittle 1981); the original literature citations are provided in Table 4.

Two other sequences were compared in this study in an effort to settle disputed common-ancestry claims. In one case the on-

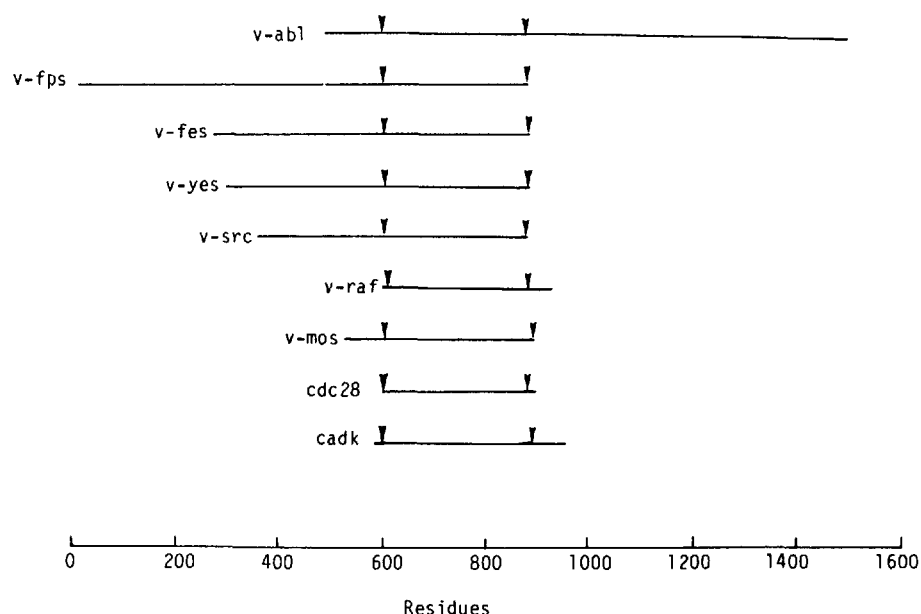


Fig. 2. Diagrammatic alignment of tyrosine kinase-like sequences. The segments used for comparison are set off by arrows. (See also Table 4)

Table 4. The nine tyrosine kinase-type sequences compared in this study

Designation	Source	Full sequence length (residues)	Portion compared (residues) ^a	Length compared (residues)
v-src	Avian (Schwartz et al. 1983)	526	254-526	273
v-yes	Avian (Kitamura et al. 1982)	592	318-592	275
v-abl	Murine (Reddy et al. 1983)	1009	110-389	280
v-fes	Feline (Hampe et al. 1982)	609	331-609	279
v-fps	Avian (Shibuya and Hanafusa 1982)	873	595-873	279
v-raf	Murine (Rapp et al. 1983)	323	13-297	285
v-mos	Murine (Van Beveren et al. 1981)	374	79-374	296
cdc28	Yeast (Lorincz and Reed 1984)	298	1-286	286
cadk	Bovine (Shoji et al. 1981)	350	19-297	279

^a Only the portions of the sequences that correspond to the sequence of cdc28 were used for the comparisons

cogene sequence v-rel, which has been suggested as possibly belonging to the tyrosine kinase oncogene family (Stephens et al. 1983), was examined and its relationship assessed. In another case, the sequence of rhodanese was reconsidered to see how closely the alleged relationship between the amino- and carboxy-terminal domains (Keim et al. 1981) is borne out by the various schemes.

Results

Alignments

Binary comparisons of *closely* related sequences, whether from the globin family or the tyrosine ki-

nase-like family, gave equivalent alignments regardless of which of the four methods was used. This is to say that the percentage identity and the number of gaps were virtually constant for the four approaches when the degree of similarity was greater than 30%. When the average percentage identity drops below 30%, however, considerable variation may occur in both the exact percentage identity itself and the number and positions of gaps in alignments generated by the various procedures. Careful scrutiny of individual alignments revealed that although large portions of the alignments generated by the

Table 5. Statistical significance of alignment scores obtained for the nine globin sequences by each of the four different scoring procedures^a

Comparison	UM	LOM	SG	GC
hbhu/hghu ^b	57.5	40.4	38.2	45.4
hbhu/hahu ^b	22.0	19.4	19.7	21.8
hbhu/myhu	8.8	9.5	9.5	7.6
hbhu/heha	5.3	5.9	5.4	5.1
hbhu/mycr	4.3	8.0	6.9	6.0
hbhu/hety	4.4	4.0	2.8	2.7
hbhu/haew	2.8	4.6	4.0	3.8
hbhu/gpfb	1.3	7.8	7.3	4.2
hghu/hahu ^b	22.5	19.0	19.0	19.4
hghu/myhu	8.1	11.7	12.1	8.3
hghu/heha	5.8	6.4	5.8	5.6
hghu/mycr	2.6	9.3	7.8	4.9
hghu/hety	5.3	4.9	2.9	3.1
hghu/haew	3.1	5.2	4.4	3.4
hghu/gpfb	3.0	8.4	6.7	4.3
hahu/myhu	12.9	12.4	11.5	11.0
hahu/heha	6.8	7.5	6.6	5.8
hahu/mycr	3.2	9.3	7.5	4.9
hahu/hety	5.4	6.6	6.2	6.2
hahu/haew	0.2	7.6	5.7	3.4
hahu/gpfb	0.4	7.8	6.4	1.5
myhu/heha	3.3	3.4	2.9	1.9
myhu/mycr	3.9	7.1	7.5	8.0
myhu/hety	1.2	2.9	3.6	1.8
myhu/haew	-0.5	3.8	3.9	2.3
myhu/gpfb	1.2	5.0	4.8	2.3
heha/mycr	2.4	4.7	3.9	2.6
heha/hety	-0.3	3.7	4.6	1.8
heha/haew	3.0	3.5	2.9	2.8
heha/gpfb	1.7	2.8	2.9	1.5
mycr/hety	3.1	6.2	3.6	3.9
mycr/haew	2.6	4.9	5.0	3.2
mycr/gpfb	-0.3	6.8	5.7	3.5
hety/haew ^b	15.8	15.4	13.8	13.9
hety/gpfb	0.2	3.2	2.9	2.2
haew/gpfb	2.2	6.4	6.6	2.9
Mean	6.3	8.2	7.5	6.5
Mean ($<30\%$ ID)	3.4	6.3	5.6	4.1

^a Results are expressed as the number of standard deviations (SD) that the authentic alignment score is above (or below) the mean of alignment scores obtained for randomized sequences of the same lengths and compositions as the authentic sequences. One hundred jumbled comparisons were performed for each pair of sequences

^b Comparisons in which the percentage identity is more than 30; the lower set of mean values shown does not include these data

Table 6. Statistical significances of alignment scores for binary comparisons of the nine tyrosine kinase-like sequences^a

Comparison	UM	LOM	SG	GC
v-src/v-yes ^b	82.7	52.9	52.1	71.1
v-src/v-abl ^b	47.2	41.3	31.7	31.1
v-src/v-fes ^b	40.6	25.7	23.8	21.7
v-src/v-fps ^b	34.2	25.4	27.2	29.0
v-src/v-raf ^b	17.9	16.2	16.5	15.7
v-src/v-mos	14.3	10.2	11.3	11.0
v-src/cdc28	4.7	9.2	9.6	6.9
v-src/cadk	8.1	9.4	9.5	8.0
v-yes/v-abl ^b	44.8	29.8	33.9	45.2
v-yes/v-fes ^b	36.5	31.7	28.0	38.4
v-yes/v-fps ^b	32.0	26.7	29.9	30.0
v-yes/v-raf ^b	21.7	19.5	19.7	24.8
v-yes/v-mos	13.3	12.8	14.4	9.8
v-yes/cdc28	5.7	8.4	6.9	8.6
v-yes/cadk	4.9	12.7	9.7	6.9
v-abl/v-fes ^b	42.5	33.4	32.3	29.8
v-abl/v-fps ^b	38.0	41.6	35.1	33.4
v-abl/v-raf	14.4	18.1	14.8	16.3
v-abl/v-mos	6.4	10.7	11.0	9.6
v-abl/cdc28	8.1	8.2	8.5	9.3
v-abl/cadk	10.1	12.2	12.6	9.7
v-fes/v-fps ^b	86.7	58.8	59.7	66.1
v-fes/v-raf	14.3	18.5	18.8	15.8
v-fes/v-mos	10.1	11.5	12.1	13.0
v-fes/cdc28	5.8	7.2	8.9	7.5
v-fes/cadk	8.7	9.6	9.1	6.5
v-fps/v-raf	15.9	15.4	15.6	13.5
v-fps/v-mos	10.6	13.8	10.7	9.7
v-fps/cdc28	8.8	7.4	9.0	8.5
v-fps/cadk	9.6	10.2	9.6	9.8
v-raf/v-mos	8.9	10.7	9.2	9.7
v-raf/cdc28	6.4	6.4	9.4	9.0
v-raf/cadk	4.9	4.7	5.3	5.5
v-mos/cdc28	9.5	9.9	7.5	8.8
v-mos/cadk	9.1	5.5	7.1	6.3
cdc28/cadk	7.9	7.4	7.7	8.4
Mean	20.7	18.1	17.7	18.5
Mean ($<30\%$ ID)	9.2	10.4	10.3	9.5

^a Results are expressed in terms of standard deviations. Thirty-six randomized comparisons were used in each case. All of the relationships were highly significant by all four procedures

^b Comparisons in which the percentage identity is more than 30; the lower set of mean values shown does not include these data

nificance measurements of the alignments produced by each of the four methods, as assessed in the manner described in the Methods section, are presented in Table 5. If a cutoff criterion of $+3.0$ SD is regarded as validation of homology, then the Dayhoff LOM and the SG procedures proved superior to the UM and the GC approaches. In the 36 different comparisons of the nine globins, all but two of the LOM significance scores were greater than $+3.0$ SD. Similarly, in only six cases did the SG method fail to produce a value above this cutoff score, and these missed only by a narrow margin.

Sensitivity of the Alignment Procedures

Globins. It was assumed a priori that all of the nine globin sequences are indeed homologous. The sig-

different methods for a given comparison are identical, major differences occur in the most dissimilar sections of two distantly related sequences.

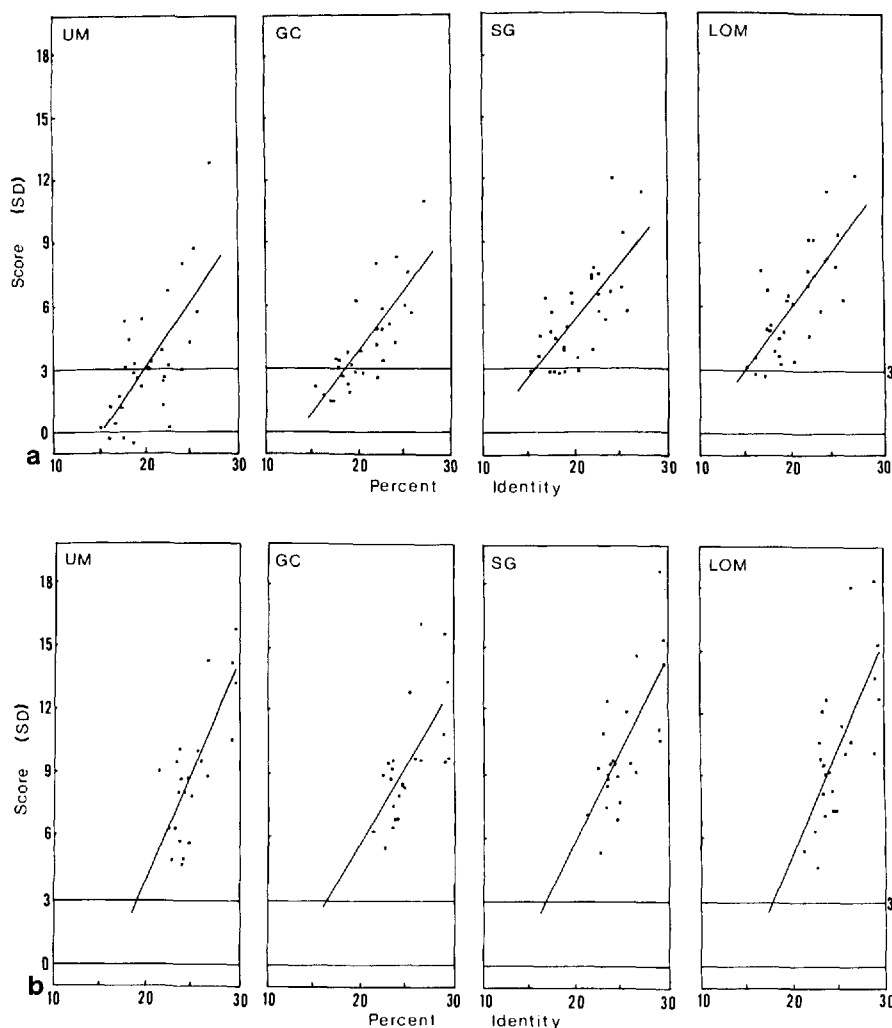


Fig. 3a, b. Comparison of four different alignment approaches with regard to statistical significance for each binary comparison of **a** globins and **b** tyrosine kinase-like sequences. Only values for those pairs of sequences that are less than 30% identical are plotted. The percentage identity used for any pair of sequences was determined by averaging the percentage identities found using the four methods. Lines were fitted by linear regression analysis

On the other hand, although the UM scoring system produced the overall highest scores for closely related sequences, it did not achieve the accepted level of significance in 9 of 36 pairwise comparisons. Alignments based on minimization of the number of base changes (GC method) between two sequences failed 12 times. Interestingly, nine of the failures were common to both approaches (see Table 5).

Because the value of weighting ought to be felt most in comparisons of the least similar sequences, more attention was given to those alignments involving pairs that were less than 30% identical (as determined by averaging all four percentage identities). Graphical analysis of these data (Fig. 3a) confirmed that the LOM method was the most effective by the criterion of the most distant relationship (as gauged by the smallest percentage identity) for which the +3.0 cutoff was exceeded. The UM approach was the least effective.

Tyrosine Kinase-Type Sequences. For the tyrosine kinase-type sequences, there was very little dif-

ference among the four methods (Table 6). Overall, the average scores obtained with the UM approach were higher than those from any of the other three scoring systems, although this result is biased to a degree by very high scores for the most similar sequences. Of the four scoring systems, the UM approach gave the highest value in 11 of 36 authentic comparisons; it was followed by the SG procedure (9 of 36). The LOM and GC methods each produced the highest score eight times. For the most distant comparisons—those for which the percentage identity was less than 30—the SG approach scored the highest in 9 of 24 analyses. The LOM procedure scored best seven times, the GC five, and the UM three.

Graphical analysis (Fig. 3b) of the alignment significance measures from comparisons of sequences with average percentage identities less than 30 confirmed that all were quite similar in effectiveness. The three criteria considered were (a) intersection point for +3.0 SD; (b) slope, which ought to reflect increasing contribution of weighting as distance increases; and (c) standard error, as an index of general

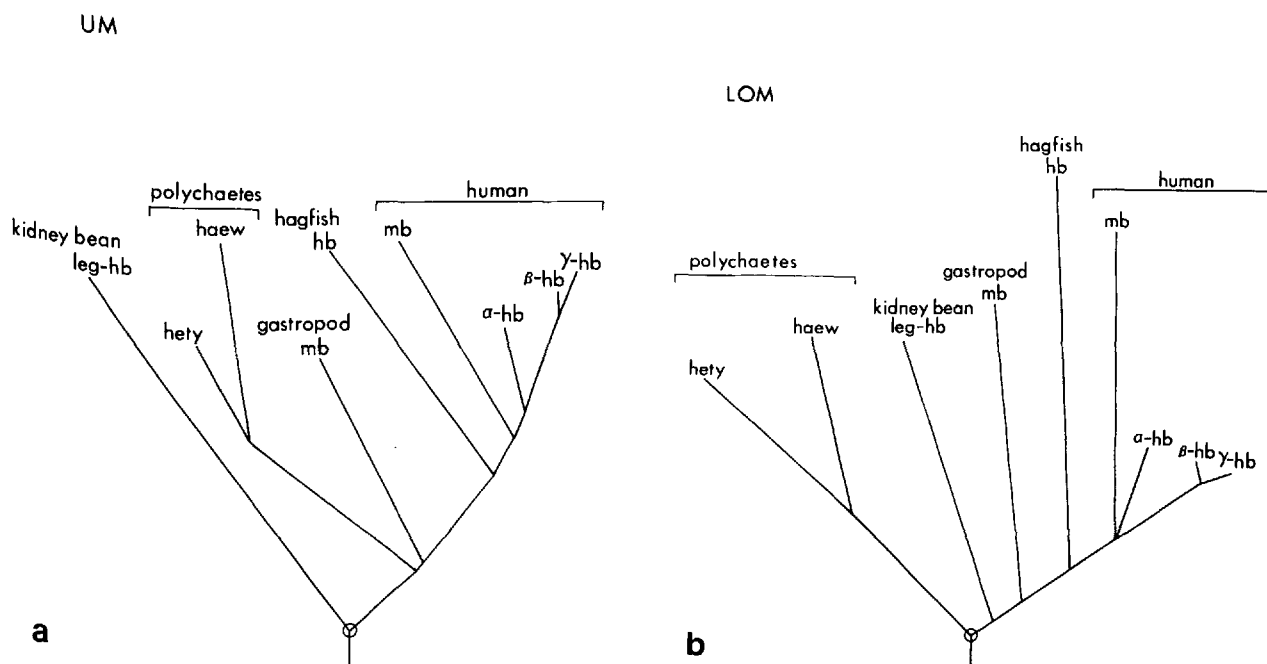


Fig. 4a, b. Globin phylogenetic trees determined from **a** UM comparisons and **b** LOM comparisons. Note the difference in branching topology for the earthworm globins and leghemoglobin between the two trees

reliability. In this set of comparisons the LOM method was not significantly better than the other three.

Phylogenetic Trees

Globins. Phylogenetic trees were constructed using both the UM and LOM approaches. In both cases the branching orders for the various globins were similar, with the possible exception of the relative positioning of the two earthworm sequences and leghemoglobin (Fig. 4), although there is a somewhat arbitrary aspect to the positioning of the roots in each case. Because the data from which the UM tree were calculated included a number of values that clearly did not achieve statistical significance, it might be expected that the percentage standard deviation (PSD) for the UM tree would be substantially higher than that for the LOM tree. Alternative trees were examined in each case in order to find the minimum PSD. The PSDs between branch lengths of these newly created trees and the original difference scores from which they were constructed were calculated. The branch lengths differ significantly for the trees derived from the two different sets of data (Fig. 4).

Tyrosine Kinase-Type Sequences. Phylogenetic trees were constructed for the tyrosine kinase-type sequence family using the data from the LOM approach and the UM approach. The branching orders

were virtually identical (Fig. 5). The PSDs of the LOM and UM trees were similar, as were the summed minimal branch lengths. Indeed, in this case the relative branch lengths of the two trees differed only slightly.

V-Rel: A Special Case

The sequence of the transforming factor from the turkey lymphatic leukemia virus (v-rel) was recently reported (Stephens et al. 1983). The suggestion was made in that report that v-rel might be homologous to the tyrosine kinase family, even though the oncogene itself has not yet been shown to have this activity. The relationship was tenuous enough that the authors posed the possibility in the most cautious of terms. Indeed, one of us (R.F.D.), responding to an earlier personal query as to whether the sequence was related to the tyrosine kinase family, had concluded that it was not. Given the different results obtained with the four approaches when applied to various proteins, we decided to reexamine the v-rel sequence and compare it carefully with these selected members of the tyrosine kinase family.

When a 272-residue segment of the 503-residue sequence was compared with corresponding segments from the other oncogenes, none of the four methods yielded consistently significant scores (Table 7). If, on the other hand, smaller segments were used, as was suggested by Stephens et al. (1983),

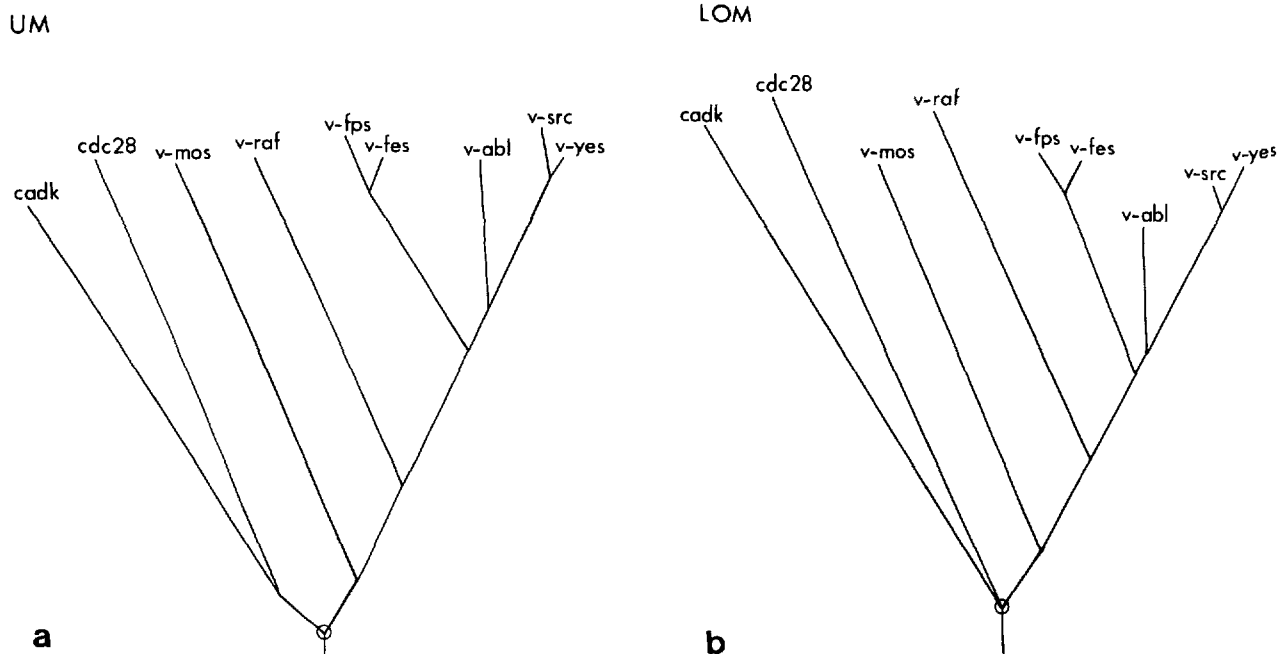


Fig. 5a, b. Phylogenetic trees for tyrosine kinase-like sequences as determined by a the UM method and b the LOM method. The topologies of the trees are essentially the same and the limb lengths differ only slightly

Table 7. Statistical significances of comparisons between v-rel and the seven tyrosine kinase-like sequences^a

Comparison	UM			LOM			SG			GC		
	A	B	C	A	B	C	A	B	C	A	B	C
v-rel/v-src	-0.7	3.7 ^b	1.9	2.0	5.6 ^b	3.8 ^b	1.3	4.0 ^b	3.3 ^b	-0.5	4.8 ^b	2.8
v-rel/v-yes	0.0	1.9	0.4	3.0 ^b	7.1 ^b	4.0 ^b	2.1	4.4 ^b	2.7	-0.5	3.8 ^b	1.9
v-rel/v-abl	1.9	-0.3	2.6	0.6	1.0	2.6	-1.2	0.2	1.5	-0.4	-0.4	1.7
v-rel/v-fes	0.5	1.7	1.0	2.8	1.5	2.5	0.2	0.6	0.9	0.1	1.1	1.3
v-rel/v-fps	0.7	0.8	1.6	2.1	1.2	2.4	-0.5	0.0	1.1	0.1	0.8	1.5
v-rel/v-raf	-0.8	-0.6	-0.7	1.4	-0.9	0.2	-1.0	0.4	-0.8	0.1	1.0	-1.6
v-rel/v-mos	1.0	-0.2	2.4	2.0	1.5	5.5 ^b	1.3	0.2	4.8 ^b	3.3 ^b	0.4	4.7 ^b
Mean	0.4	1.0	1.3	2.0	2.4	3.0	0.3	1.4	1.9	0.2	1.8	1.8

^a Several different segments of these proteins were compared. In the first of these comparisons (column A), the segments of tyrosine kinase-like sequences listed in Table 4 were compared with a 272-residue portion of v-rel (residues 129–400). In a second set of comparisons (column B), a short segment of v-rel (residues 129–178), originally compared with v-src by Stephens et al. (1983), was compared with the corresponding portions of the tyrosine kinase-like sequences. Finally, a 122-residue segment of v-rel (residues 279–400) was compared with the corresponding segments of the tyrosine kinase-like sequences (column C). This portion of v-rel had also been compared with v-src by Stephens et al. (1983). Thirty-six jumbled comparisons were employed in each analysis

^b Value statistically significant by the criterion employed

some significant relationships were revealed by all four methods (Table 7). One has to be very cautious in assessing data when a preselection of this sort is made, since the counterweight of preselection is not reflected in the significance score. We return to this point in the Discussion section.

Rhodanese Domains

In a study conducted several years ago, x-ray crystallography of rhodanese revealed it to have two equivalent domains made up of the amino-terminal

and carboxy-terminal halves of the polypeptide chain, but sequence comparisons did not immediately reveal any evidence of homology (Ploegman et al. 1978). Keim et al. (1981) subsequently examined the sequence more carefully and concluded there was indeed evidence for common ancestry if the genetic code was used in conjunction with an alignment based on the crystal structure as a basis for comparison. We reexamined the situation using all four approaches to compare the amino- and carboxy-terminal halves of the sequence without regard to the x-ray structure. The significance scores ob-

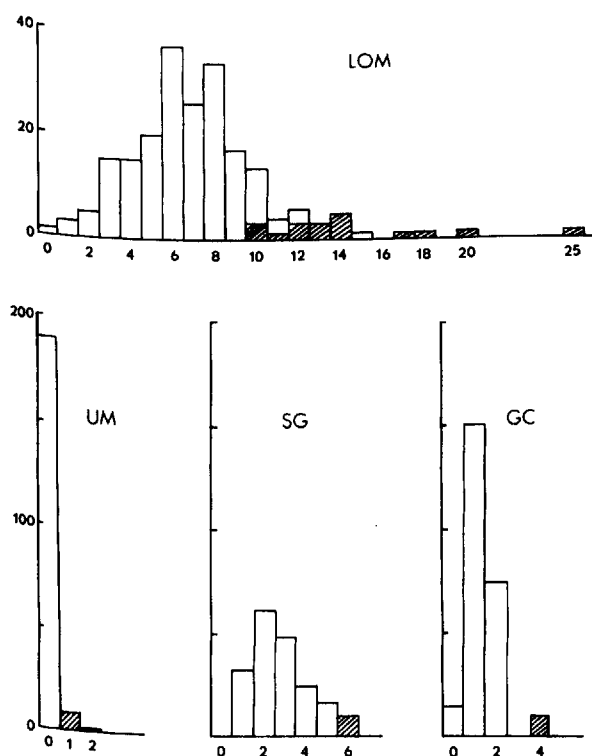


Fig. 6. Distribution of weights in the four different alignment schemes that were used in this study. The shaded areas indicate scores for identities. Because in each system only a single value is given for each of the 190 possible amino acid interchanges, the twenty identities were tallied as 0.5 each

tained were 4.1, 2.6, 2.2, and 1.5 SD for the LOM, UM, SG, and GC approaches, respectively.

Discussion

Weighting Schemes

Weighting schemes are intended to enhance the detection of common ancestry between distantly related sequences. The underlying assumption is that evolutionary interchanges are more likely between amino acids that have similar structures or that can exchange by single-base substitution. The intent is to ascertain that a given sequence resemblance falls outside the realm of reasonable chance, whether that resemblance be due to common ancestry or structural convergence.

There are two general approaches to assigning weights. In one, the weighting is made on the basis of the genetic code and, in some cases, ad hoc considerations of what constitute similar structural properties for the amino acids. In the other approach, the most sophisticated and elegant version of which is the Dayhoff LOM, the weighting is based on the observed frequency of amino acid interchanges in related proteins. This approach has also been used by McLachlan (1971).

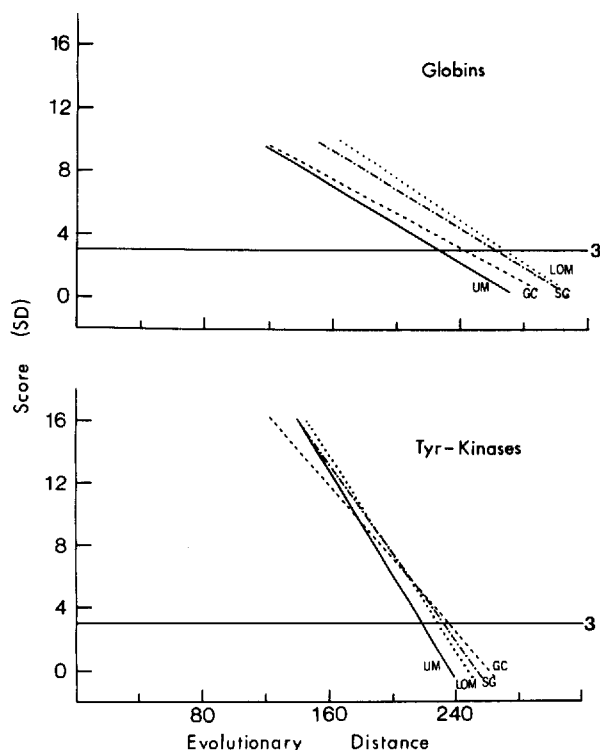


Fig. 7. Comparison of four different alignment approaches with regard to statistical significance as a function of evolutionary distance. The evolutionary distance was taken to be the negative log (base e) of the average percentage identity adjusted for 10% random identity. The percentage identity used for any particular pair was the average of the four percentage identities obtained with the four methods

The degree of weighting varies from system to system. Thus, McLachlan (1972) originally devised a very conservative weighting scheme that differed from the two-value system of the UM approach only in that it gives some modest weight to 20 of the 190 interchanges and a very small weight to 25 others. In the weighted SG scheme that we used in this study, the weighting is considerably more generous (Fig. 6). Similarly, the empirical LOM method of Dayhoff gives some weight to all of the possible amino acid interchanges except one (tryptophan/cysteine). It is interesting to note that in the LOM scheme it is just as "rewarding" to have asparagine matched with aspartic acid or histidine as it is to have two asparagines (Table 1).

The question arises, is weighting effective? The answer is apparently yes, although in any particular situation it may not be obvious. As early as 1972, Dayhoff concluded, on the basis of a small number of comparisons of different approaches, that the LOM was superior to other schemes. In our own casual experience this seemed not to be the case, but the results of the extensive survey reported here confirm that—on the average—weighting can help in establishing distant relationships.

We may then ask which weighting scheme is most effective. The LOM method was the best in the globin comparisons, as shown by the fact that the intersection point for significance occurs at the greatest evolutionary distance (Fig. 7). It was, however, closely followed by the SG method. The good performance of the LOM method may be due in part to the fact that hemoglobin data contributed disproportionately to the early tabulation of observed species changes. In contrast, the LOM method was not any better, on the average, in establishing the relationships of the tyrosine kinase-like sequences. The LOM was the best in bringing out the sequence similarity between the two homologous domains of rhodanese.

Is V-Rel Related to the Tyrosine Kinase Family?

The comparisons of full-length or nearly full-length sequences of v-rel with the tyrosine kinase-like oncogene sequences did not yield significant alignment scores. On the other hand, selected segments did score at levels implying an authentic relationship. This use of "iceberg segments"—those whose statistical significance still "protrudes above the surface" of time-worn change—is a legitimate approach when there is other evidence suggesting a relationship. In this case the other evidence is that v-rel is a genuine oncogene that can transform appropriate cells, as do the other principles listed in this group.

Interestingly, the segments of v-rel are not consistently related to the same members of the group. Thus, the segment corresponding to residues 129–178 has a reasonable correspondence to v-src and v-yes, the latter two being themselves quite similar. On the other hand, the section of v-rel comprising residues 279–400 resembles v-mos more closely than it does any of the others. This implies that v-rel may have been generated by something more than simple amino acid replacements along a single line. Rather, it seems to reflect a chromosomal aberration or transposition, perhaps an unequal crossing-over between the host genes corresponding to v-src on the one hand and v-mos on the other.

Phylogenetic Trees

Some heed must be given to the biological significance of the trees generated in this study. It should be made clear that the positioning of the roots is somewhat arbitrary. If the topology of the LOM tree for globins is accepted as drawn, then it is apparent that leghemoglobin must have branched off at about the same time as the divergence of lower invertebrates and the line leading to vertebrates. If this is true, then it could have involved the "lateral trans-

fer" of genetic information, since the divergence of plants or bacteria obviously predated that occasion. On the other hand, if the divergence of leghemoglobin is situated at a more distal point, as is drawn for the UM tree, a result consistent with a more conventional origin for leghemoglobin is obtained. A similar tree was derived by others (Goodman et al. 1974) using the equivalent of the GC approach.

The oncogene trees are also quite revealing, showing there to be absolutely no question that the previously reported relationships among cyclic AMP-dependent kinase (cadk), the yeast cell division cycle factor cdc28, and the seven oncogene sequences studied are valid. The point is underscored by a casual consideration of the large number of positions where all nine sequences have the same amino acids (Fig. 8).

The number of genes actually involved here is not altogether clear, however. For example, v-fps and v-fes are almost certainly products of the "same" gene that has been pirated on independent occasions by different retroviruses, with the differences in sequence reflecting the species difference between the avian and feline host sequences. On the other hand, the difference between v-fps and v-src, both of which stem from avian viruses, probably indicates two different but related host genes. The same is likely true of v-src and v-yes. The question becomes, to which of these two or three genes does the murine-generated v-abl correspond? Future trees that include the sequences of more recently reported tyrosine kinase oncogenes, such as v-fms (Hampe et al. 1984), may help to clarify these relationships.

Normalized Alignment Scores as Guides to Significance

It would be useful if a rough index of significance could be obtained from an alignment score directly and without recourse to the extensive computer time involved in "jumbling" tests. The data we have collected in this study indicate that such a gauge is indeed possible, providing suitable regard is taken of the lengths of the sequences being compared (Fig. 9). For example, in UM alignments, a NAS greater than 175 is a good indication of a genuine relationship. The longer the sequences involved, the more reliable the assurance, of course, and for sequence lengths on the order of those of the tyrosine kinase-like proteins studied here, even a NAS as low as 150 is a clear indication of common ancestry. The intersections with the +3.0 line for the scores generated in the other three systems also offer reliable guides to significance (Fig. 9).

The NAS-significance cutoff points in Fig. 9 can also be used in conjunction with "eyeball align-

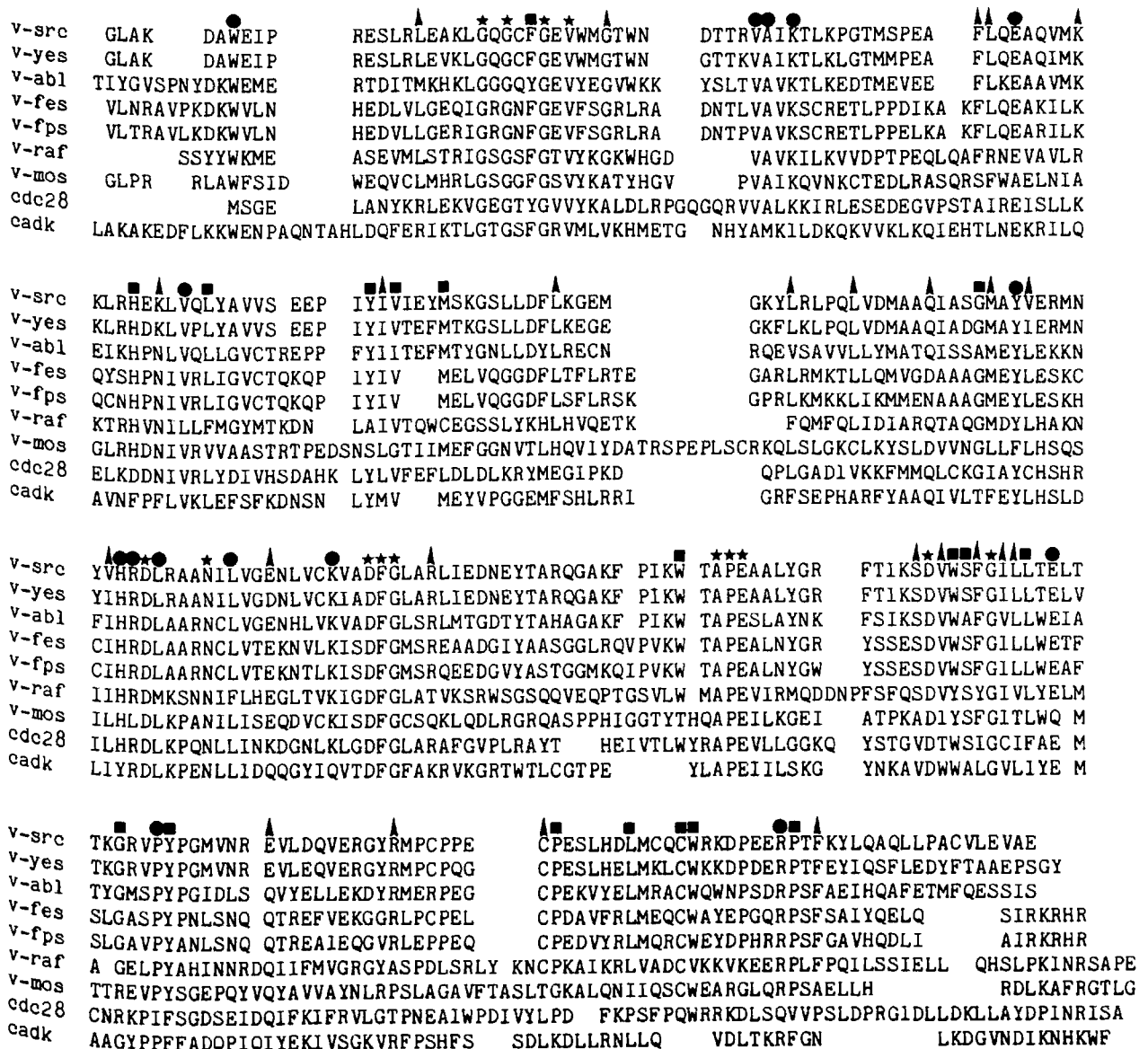


Fig. 8. Overall alignment of the seven tyrosine kinase-like oncogene sequences and those of yeast *cdc28* and bovine heart cyclic AMP-dependent kinase. Stars denote locations where all nine residues are identical; circles, eight of nine; squares, seven of nine; and arrowheads, six of nine.

ments" made without computer assistance. The most straightforward application would make use of the UM scoring system, since in that case only identities and gaps need be counted. The other methods can be used in such a setting, however, especially if a programmable calculator is used. Recourse to these guidelines might reduce the number of unsubstantiated claims of "homologous" protein sequences.

Conclusion

There has been a growing interest in exploring evolutionary relationships among protein sequences in

the past few years. The number of known amino acid sequences has grown enormously and increased the chances of identifying related proteins. The difficulties involved in distinguishing common ancestry or structural convergence from strictly chance resemblance are becoming well appreciated. What each investigator wants to know is, what is the best way to align two amino acid sequences to see if they share common ancestry? In contrast to what one of us has implied in the past (Doolittle 1981), the answer may be to use a weighted matrix. Although there is no certainty that weighting will help in any particular situation, it is seldom a handicap. The best weighting schemes take account of both genetic

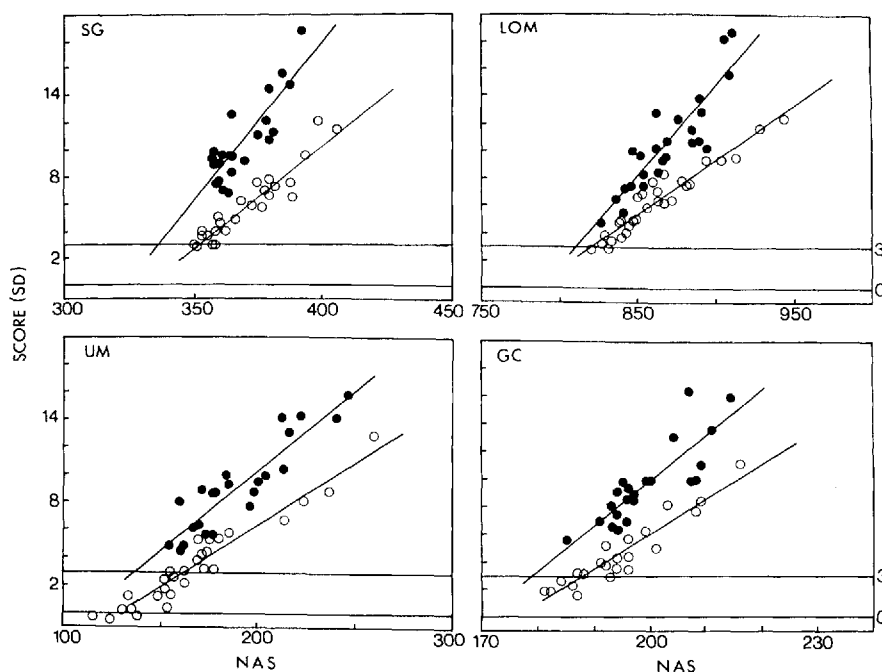


Fig. 9. Plots of significance scores (in terms of standard deviations) vs normalized alignment scores (NAS). For any of the four systems, $NAS = 100 \times \text{alignment score} / \text{number of residues in the shorter sequence}$. Open circles represent data for globins; solid circles, those for tyrosine kinase-like sequences

likelihood and structural similarities between amino acids.

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